

Smart Traffic Lights Using Local AI for Emergency Vehicle Route Clearance

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Abstract—Smart traffic-light systems are increasingly being studied to improve traffic-flow efficiency under changing road conditions. Traditional traffic signals generally operate using predefined timing schemes and may not respond efficiently to sudden changes in traffic density. This paper proposes a local-AI-based smart traffic-light architecture that uses street cameras and smart-pole computers to process traffic information close to the road. The proposed system is designed to reduce dependence on continuous cloud communication and provide faster traffic decisions. An additional objective is to detect emergency vehicles and coordinate nearby traffic signals to create a clear route. The proposed architecture consists of street cameras, local processing units, a city cloud server, and traffic-light controllers. The paper discusses the system workflow, mathematical model, algorithm, comparison with conventional approaches, and possible future improvements.

Index Terms—Smart Traffic Lights, Local AI, Traffic Signal Control, Machine Vision, Reinforcement Learning, Emergency Vehicle Detection, Intelligent Transportation Systems

I. INTRODUCTION

Smart traffic-light systems are increasingly being studied to improve traffic-flow efficiency under changing traffic conditions. Reinforcement learning has been investigated as a method for adaptive traffic signal control [1].

Machine-vision-based approaches have also been explored for obtaining traffic-flow information and dynamically adjusting traffic-light operation [2].

Traditional traffic lights generally operate using predefined timing schedules. These systems are simple and inexpensive, but their fixed operation may not respond effectively when traffic density changes unexpectedly.

A local-AI-based traffic-light system can process traffic information closer to the location where it is collected. Instead of continuously transmitting complete camera streams to a remote server, a local processing unit can extract relevant information and communicate a shorter decision to the traffic-light controller.

The proposed work focuses on a smart traffic-light architecture in which street cameras monitor traffic, smart-pole computers perform local processing, and traffic-light controllers use the resulting information to adjust signal operation.

An important application considered in this work is emergency vehicle route clearance. When an ambulance approaches a sequence of intersections, the system can identify the emer-

gency vehicle and coordinate nearby signals to provide a clearer route.

II. RELATED WORK

Wei et al. proposed IntelliLight, a reinforcement-learning approach for intelligent traffic-light control that uses traffic information to dynamically select signal actions [?].

Zheng et al. proposed a phase-competition approach for traffic signal control, where competing traffic phases are considered when selecting signal actions [3].

For coordinated traffic control across multiple intersections, CoLight investigated network-level cooperation between traffic signals [4].

Large-scale decentralized traffic signal control has also been studied using deep reinforcement learning [5].

Machine-vision-based adaptive traffic-light control has also been investigated for obtaining traffic information from camera systems and optimizing traffic-light operation [2].

The reviewed studies demonstrate the potential of machine learning, reinforcement learning, machine vision, and cooperative traffic control for intelligent transportation systems. However, the proposed work focuses specifically on combining local processing with emergency vehicle route clearance.

III. PROBLEM STATEMENT

Conventional traffic signals often operate according to fixed schedules and may not respond effectively to sudden traffic congestion or emergency situations.

Cloud-based intelligent traffic systems can process large amounts of information but may depend on network connectivity and communication between road-side devices and remote servers.

Therefore, there is a need for a traffic-light system that can process important traffic information locally and respond quickly to emergency vehicles.

The proposed system addresses this problem by using street cameras, local smart-pole computers, and traffic-light controllers to support adaptive traffic management and emergency route clearance.

IV. OBJECTIVES

The main objectives of the proposed system are:

- To monitor traffic conditions using street cameras.

- To process traffic information locally using smart-pole computers.
- To reduce dependence on continuous cloud communication for immediate traffic decisions.
- To detect emergency vehicles and identify their direction of travel.
- To coordinate nearby traffic signals to create a clear route for emergency vehicles.
- To improve the responsiveness of traffic-light control under changing traffic conditions.

V. METHODOLOGY

The proposed methodology consists of four major stages: traffic monitoring, local processing, cloud-level storage and analysis, and traffic-light control.

First, street cameras capture information about vehicles and traffic conditions. The camera information is then provided to smart-pole computers for local processing.

The local processing unit analyzes the incoming information and determines the appropriate traffic decision. Selected information can also be forwarded to the city cloud server for storage and wider analysis.

Finally, the generated decision is communicated to the traffic-light controller.

The overall information flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

For emergency situations, the local processing unit can identify an emergency vehicle and communicate the required signal changes to nearby intersections.

A. System Flowchart

The overall workflow of the proposed system is shown in Fig. 1.

VI. SYSTEM ARCHITECTURE

The proposed system consists of four main components: street cameras, smart-pole computers, a city cloud server, and traffic-light controllers.

A. Street Cameras

Street cameras capture information about the traffic conditions at an intersection. The captured information may include vehicle presence, traffic density, and possible emergency vehicles.

B. Smart Pole Computers

Smart-pole computers perform local processing of camera information. The objective is to perform time-sensitive traffic analysis close to the road instead of depending entirely on a remote cloud server.

C. City Cloud Server

The city cloud server can be used for storing selected traffic information and performing wider traffic analysis. It is not required for every immediate traffic decision.

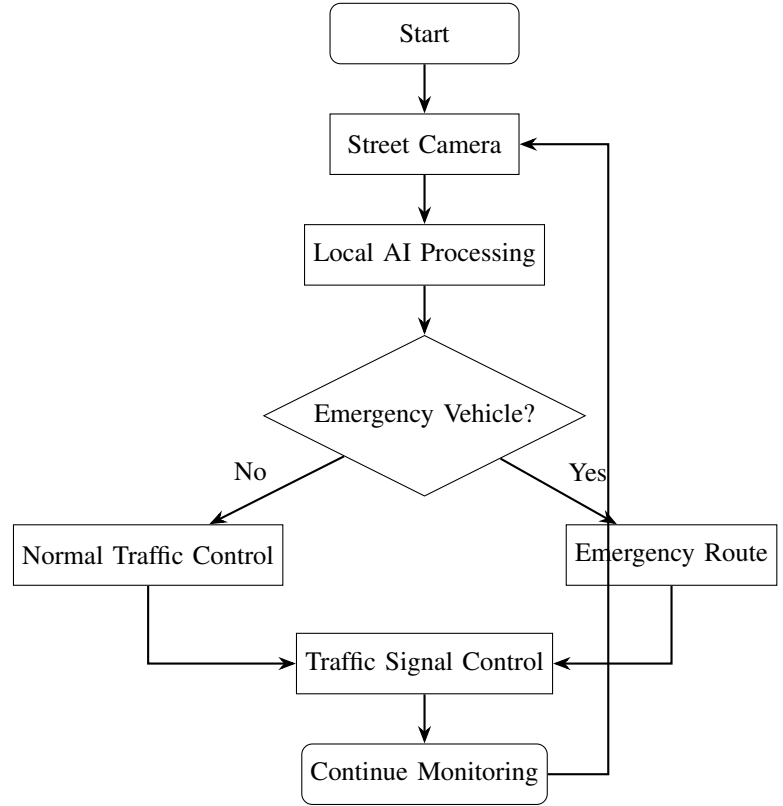


Fig. 1: Flowchart of the Proposed Local-AI Traffic System

D. Traffic-Light Controller

The traffic-light controller receives the decision from the local processing system and changes the signal state accordingly.

E. Architecture Diagram

The conceptual architecture of the proposed system is shown in Fig. 2.

VII. RESEARCH GAP

Existing traffic systems can work effectively for regular traffic patterns, particularly when traffic conditions are predictable.

However, emergency situations can require a faster and more coordinated response. A traffic system that can detect an approaching ambulance and coordinate nearby traffic signals may help reduce unnecessary waiting at intersections.

The proposed work therefore focuses on the combination of local traffic processing and emergency vehicle route clearance.

VIII. MATHEMATICAL MODEL

Let the traffic density at an intersection be represented by the number of detected vehicles divided by the observation area:

$$D = \frac{N_v}{A} \quad (1)$$

architecture.png

Fig. 2: Conceptual Architecture of the Proposed System

where N_v represents the number of detected vehicles and A represents the observation area.

The accuracy of an emergency-vehicle detection model can be represented as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

where TP represents True Positive, TN represents True Negative, FP represents False Positive, and FN represents False Negative.

Precision is calculated as:

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

Recall is calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

IX. ARRAY AND MATRIX REPRESENTATION

Traffic information collected from an intersection can be represented as a feature matrix:

$$A = \begin{bmatrix} v_1 & d_1 & s_1 \\ v_2 & d_2 & s_2 \\ v_3 & d_3 & s_3 \end{bmatrix} \quad (5)$$

where v represents the number of vehicles, d represents traffic density, and s represents the signal state.

The feature matrix can be provided to the local processing model for determining the appropriate traffic-control action.

Algorithm 1 Local-AI Traffic Signal Control

- 1: Initialize the traffic monitoring system.
- 2: Capture traffic information using street cameras.
- 3: Send camera information to the local processor.
- 4: Detect vehicles and estimate traffic conditions.
- 5: Check for an emergency vehicle.
- 6: **if** emergency vehicle is detected **then**
- 7: Determine the required route.
- 8: Coordinate nearby traffic signals.
- 9: Provide a clear route for the emergency vehicle.
- 10: **else**
- 11: Apply normal traffic signal control.
- 12: **end if**
- 13: Store selected traffic information in the cloud.
- 14: Continue monitoring.

TABLE I: Comparison of Traffic Signal Approaches

Feature	Timed	Cloud AI	Local AI
Adaptive Control	Low	High	High
Internet Dependence	Low	High	Lower
Local Processing	No	Limited	Yes
Emergency Support	Limited	Possible	Proposed
Response	Fixed	Network Dependent	Local

TABLE II: Example Experimental Result Format

Method	Accuracy	Precision	Recall
Method 1	—	—	—
Method 2	—	—	—
Proposed	—	—	—

X. PROPOSED ALGORITHM

The basic operation of the proposed system is given in Algorithm 1.

XI. SYSTEM COMPARISON

The proposed local-AI approach can be conceptually compared with conventional timed signals and cloud-based intelligent systems.

XII. EXPERIMENTAL RESULTS

For demonstration purposes, the system can be evaluated using traffic-density and emergency-vehicle detection scenarios.

Table II provides an example format for reporting experimental results. Actual values should be replaced with measurements obtained from the implemented system.

XIII. IMPLEMENTATION

The proposed system can be implemented using a camera, a local computing device, a traffic-light controller, and suitable machine-learning software.

The local processing unit receives camera information, performs vehicle detection, estimates traffic conditions, and determines the required traffic signal action.

A simplified Python representation of the calculation of an evaluation metric is shown below.

```
1 def calculate_accuracy(correct, total):
2     if total == 0:
3         return 0
4
5     accuracy = (correct / total) * 100
6     return accuracy
7
8 accuracy = calculate_accuracy(95, 100)
9 print("Accuracy:", accuracy)
```

XIV. RESULTS AND DISCUSSION

The proposed architecture provides a framework for combining local traffic analysis with adaptive traffic signal control.

Local processing can reduce the amount of information that needs to be transmitted to a remote server for immediate decisions. The cloud server can still be used for storage, monitoring, and larger-scale analysis.

The emergency vehicle component provides an additional application of the proposed architecture. When an emergency vehicle is detected, nearby signals can be coordinated to support route clearance.

The actual effectiveness of the system should be determined using experimental measurements such as detection accuracy, decision time, traffic waiting time, and emergency-route clearance time.

XV. ADVANTAGES

The major advantages of the proposed system include:

- Local processing of traffic information.
- Reduced dependence on continuous cloud communication for immediate decisions.
- Adaptive traffic signal control.
- Potential support for emergency vehicle route clearance.
- Scalable architecture for multiple intersections.
- Ability to store selected information for wider traffic analysis.

XVI. LIMITATIONS

The proposed system has several limitations.

First, reliable camera placement and adequate image quality are required for effective vehicle detection. Poor weather, occlusion, low illumination, or camera failure may affect detection performance.

Second, the local processing device must have sufficient computational capability to process traffic information within the required response time.

Third, emergency vehicle detection must be sufficiently reliable because incorrect detection may result in unnecessary changes to traffic signals.

Finally, the proposed architecture requires suitable integration with existing traffic-light infrastructure.

XVII. FUTURE SCOPE

Future work can focus on implementing the proposed architecture using real traffic-camera data and edge-computing hardware.

A more advanced emergency vehicle detection model can be developed using deep learning and computer vision. The system can also be extended to coordinate several intersections along an emergency vehicle route.

Further research can evaluate the system using real-world measurements such as traffic waiting time, vehicle throughput, emergency-route clearance time, detection accuracy, and processing latency.

XVIII. CONCLUSION

This paper presented a conceptual smart traffic-light system based on local AI processing.

The proposed architecture uses street cameras to monitor traffic conditions, smart-pole computers to perform local processing, a city cloud server for selected storage and analysis, and traffic-light controllers to implement traffic decisions.

The system additionally considers emergency vehicle route clearance as an important application. By detecting an emergency vehicle and coordinating nearby signals, the proposed approach aims to provide a more responsive traffic-management mechanism.

Future implementation and real-world experimentation are required to determine the actual performance of the proposed system.

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